Machine Learning in Time Series

Lecture 3: High-Dimensional Linear Dynamics (BVAR & Sparse VAR)

Instructor: Marcelo Villena, PhD

Spring 2026

PhD Seminar in Econometrics / Data Science
Università degli Studi di Bari Aldo Moro

Agenda

1. Seminar Overview & Roadmap

2. The Genesis of VAR:
Let the Data Speak

3. The Bayesian Solution: BVAR

4. The Machine Learning Solution: Sparse VARs

5. Lecture 3 Mini-Lab: Multivariate Forecasting

6. Concluding Remarks

7. Readings Discussion

Seminar Overview & Roadmap

The 6-Lecture Journey

Where we are

We have mastered univariate forecasting with regularization (Lecture 1) and non-linear classification/regression (Lecture 2). Today, we scale up to **Multivariate Systems**. How do we forecast an entire economy of interacting variables when $p \gg T$?

Class 1: Historical Evolution, the Prediction Problem, and Shrinkage.

Class 2: Kernel Methods and Support Vector Machines (SVM).

Lecture 3: **High-Dimensional Linear Dynamics (BVAR, Sparse VAR).**

Lecture 4: Deep Learning for Sequences (RNN/LSTM/Attention).

Lecture 5: Trees, Boosting, and Ensembles.

Lecture 6: NLP, Attention, and Causal ML.

The Genesis of VAR: Let the Data Speak

The Macroeconomic Crisis of the 1970s

Before 1980, macroeconomists relied on massive "Cowles Commission" structural models containing hundreds of equations.

To make these models mathematically solvable, researchers had to manually classify variables as strictly *endogenous* (determined by the model) or strictly *exogenous* (determined outside the model).

Christopher Sims' Critique (1980)

In his landmark paper *"Macroeconomics and Reality"*, Sims argued these manual classifications were **"incredible restrictions"**. In a highly interconnected economy, everything affects everything else.

The Solution: The Vector Autoregression (VAR). Treat *all* variables symmetrically as endogenous. Let the data discover the relationships.

The Vector Autoregression (VAR) Model

A VAR(p) models a $K \times 1$ vector of time series variables Y_t as a linear function of their own past values and the past values of all other variables in the system.

The VAR(p) Equation

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + e_t$$

A Classic 3-Variable System:

$Y_t = [\pi_t \text{ (Inflation)}, \quad u_t \text{ (Unemployment)}, \quad r_t \text{ (Interest Rate)}]^\top$

Pros of the VAR Paradigm:

- **Systemic Dynamics:** Captures complex feedback loops (e.g., Fed raises rates $\rightarrow$ unemployment rises $\rightarrow$ inflation falls).
- **Policy Analysis:** Enables Impulse Response Functions (IRFs) to trace shocks through the economy over time.
- **Forecasting:** Often outperforms large structural models in short-term out-of-sample prediction.

Clarifying the Taxonomy: VAR vs. BVAR vs. SVAR

Before we proceed to Machine Learning, we must distinguish three objects that are often confused in macroeconomic literature:

VAR: **Reduced-form forecast system.** Parameters are estimated purely via OLS. No structural claims.

BVAR: **Reduced-form + prior/shrinkage.** Adds Bayesian priors (or Ridge penalties) to fix the variance problem. Focuses on prediction, not causation.

SVAR: **Reduced-form + identification.** A Structural VAR imposes theoretical assumptions (e.g., Cholesky ordering, sign restrictions) onto the errors to isolate causal, independent economic shocks.

The Cons: The Curse of Dimensionality

Sims' "Let the Data Speak" philosophy has a fatal mathematical flaw in modern data-rich environments: **The parameter explosion.**

In a VAR(p) with K variables, the number of parameters to estimate is:

$$\text{Total Parameters} = K + pK^2 \quad (+ \text{ deterministic terms if included})$$

(or $1 + pK$ parameters per equation).

The Degrees of Freedom Crisis

If we model the macroeconomy using 20 variables ($K = 20$) with 12 months of lags ($p = 12$):

- Parameters per equation: $1 + (12 \times 20) = 241$.
- Total parameters in the system: **4,820**.

If we only have 30 years of monthly data ($T = 360$), OLS matrix inversion $(X^\top X)^{-1}$ becomes unstable, variance explodes, and out-of-sample forecasting collapses.

The Bayesian Solution: BVAR

The Minnesota Prior: The Mathematics

To solve the parameter explosion, Doan, Litterman, and Sims (1984) introduced a statistical "nudge." They observed that most macroeconomic variables behave like random walks with drift: $Y_t = c + Y_{t-1} + e_t$.

The **Minnesota Prior** formalizes this by setting the prior mean and variance for every coefficient $A_{ij}^{(k)}$ (the effect of variable j on variable i at lag k):

The Minnesota Prior: The Mathematics

Prior Means ($\mathbb{E}$)

$$\mathbb{E}[A_{ij}^{(k)}] = \begin{cases} 1 & \text{if } i = j \text{ and } k = 1 \text{ (First own lag)} \\ 0 & \text{otherwise (Cross-lags and distant lags)} \end{cases}$$

Prior Variances (Var)

The confidence we have in these prior means depends on hyperparameter weights:

- **Overall Tightness (λ_1):** How strictly we enforce the prior across the board.
- **Cross-Variable Weight (λ_2):** We are more confident that *other* variables' lags are zero compared to the variable's own lags.
- **Lag Decay (λ_3):** Variance shrinks as lag length k increases (distant past matters less).

The Empirical Wins of BVARs

Why do central banks love BVARs, and why do they often beat pure Machine Learning (like Lasso) in macroeconomics?

1. Smooth Structural Inference (IRFs):

- Sparse ML algorithms (Lasso) randomly zero-out coefficients, creating jagged, discontinuous matrices. This can complicate structural identification and make IRFs sensitive to tuning / selection.
- BVARs shrink coefficients *smoothly* toward zero but keep them in the system. This allows us to apply standard structural identification (e.g., Cholesky ordering) to generate smooth, economically theoretically consistent **Impulse Response Functions (IRFs)** even in high dimensions.

The Empirical Wins of BVARs

Why do central banks love BVARs, and why do they often beat pure Machine Learning (like Lasso) in macroeconomics?

2. Solving the Post-Selection Inference Nightmare:

- In Lasso, once a variable is selected, computing a valid p-value or confidence interval for it is mathematically extremely difficult.
- BVARs naturally generate the entire posterior distribution. We get exact Bayesian **Credible Intervals** for our forecasts and our IRFs for free, correctly accounting for parameter uncertainty.

BVAR is just Machine Learning Shrinkage!

The Bayesian Perspective:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

By setting the prior mean of most coefficients to 0, we heavily penalize the data from pulling the coefficient away from zero unless the signal is incredibly strong.

The Machine Learning Perspective: A Gaussian Prior centered at zero is mathematically identical to an L_2 **Ridge Penalty** from Lecture 1!

$$\min_{\beta} \|Y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$$

The Bridge (Giannone et al., 2015)

The tightness of the Minnesota prior is exactly analogous to tuning the λ hyperparameter. As $N \to \infty$, BVARs allow us to forecast with hundreds of variables without OLS exploding, successfully bypassing the curse of dimensionality.

The Machine Learning Solution: Sparse VARs

Why not just use BVAR? The need for Sparsity

While BVAR (Ridge) elegantly shrinks coefficients toward zero to reduce variance, it *rarely forces them to exactly zero*.

If we are using the FRED-MD dataset with 130 macroeconomic variables, do we really believe that lumber prices at lag 7 affect the CPI? A BVAR will assign a weight of 0.0001. A Machine Learning **Sparse VAR** will assign a weight of exactly 0.0.

The Sparse VAR (Lasso VAR) Formulation: Instead of a Gaussian prior (L_2), we apply the L_1 penalty across the entire VAR system:

$$\min_{A_1 \ldots A_p} \frac{1}{2T} \sum_{t=1}^{T} \left\| Y_t - \sum_{i=1}^{p} A_i Y_{t-i} \right\|_2^2 + \lambda \sum_{i=1}^{p} \|A_i\|_1$$

This performs simultaneous **model estimation** and **variable selection**.

Structured Regularization: Group Lasso

In time series, standard Lasso can be chaotic—it might keep lag 1 and 3 of unemployment, but drop lag 2. **Nicholson et al. (2020)** introduced structured penalties to respect time-series architecture.

Group Lasso Penalty:

$$\lambda \sum_{j=1}^{J} \|\beta^{(j)}\|_2$$

It shrinks pre-defined "groups" of variables together. Either the whole group stays, or the whole group goes to zero.

Lag-Based Grouping: Group all variables by lag. The algorithm will ask: "Does the 4th lag of the system matter as a whole?" If no, it zeroes out the entire A_4 matrix.

Variable-Based Grouping: Group all lags of a single variable. "Does the M2 Money Supply matter at all for this system?" If no, it drops all 12 lags of M2 simultaneously.

Practical Issues: Stability and Networks

When applying ML to VARs, we face unique time-series challenges that don't exist in standard cross-sectional regression.

1. **Stability Constraints:**
   - For a VAR to be stable for forecasting, the roots of its characteristic polynomial must lie outside the unit circle. Standard Lasso ignores this.
   - *Practical tip:* Packages like `BigVAR` can enforce stability. Always check the eigenvalues of the companion matrix!

2. **Network Interpretations (A Warning):**
   - A Sparse VAR yields a clean network of macroeconomic linkages (edges).
   - *Caution:* Interpretation of zero/nonzero edges is model-dependent and can be highly unstable across rolling windows. Treat these networks as exploratory unless formally validated.

Lecture 3 Mini-Lab: Multivariate Forecasting

Mini-Lab: The Multivariate Project

The Task

We are upgrading our Project. Instead of forecasting YoY CPI using only its own lags (AR), we will forecast a 3-variable system: **Inflation, Unemployment, and Interest Rates (Fed Funds)**.

Pipeline Requirements (using R `vars` and `BigVAR` packages):

1. **Classical OLS VAR:** Estimate a standard VAR(12). Observe what happens to OOS RMSE at horizon $h = 12$ when degrees of freedom vanish.
2. **Bayesian VAR (BVAR):** Implement a VAR with a Minnesota Prior.
3. **Sparse VAR:** Use the `BigVAR` package to fit a Lasso-VAR and a Group-Lasso VAR.
4. **Inspect the Matrix:** Look at the final coefficient matrix of the Sparse VAR to visually identify which macroeconomic indicators actually drove inflation over the test period.

Lab Results: The Multivariate Project

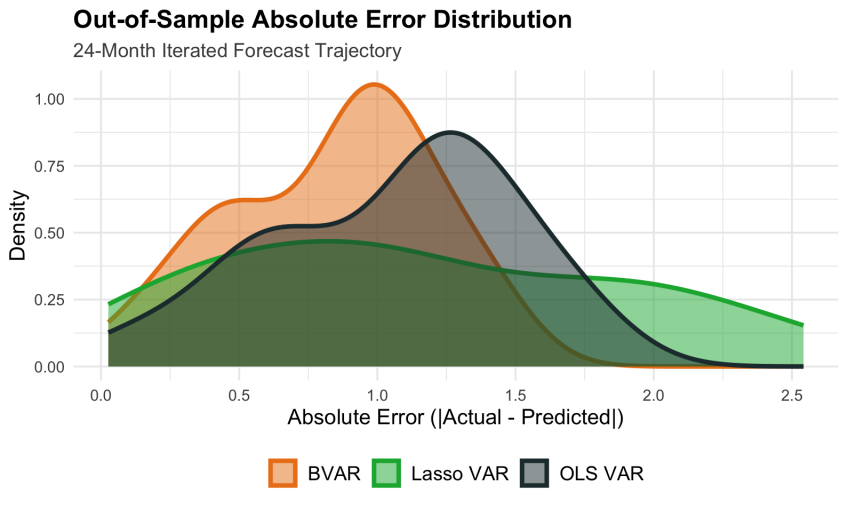
Horizon	OLS VAR Error	BVAR Error	Lasso VAR Error
$h = 1$	**0.0792**	0.1055	1.1089
$h = 3$	0.6205	0.4904	**0.2573**
$h = 6$	0.5401	**0.3977**	NA
$h = 12$	1.2921	**1.0384**	NA

Lab Results: The Multivariate Project

Takeaways: The Power of the Prior

- **Short-Term ($h = 1$):** OLS wins. The system is unbiased, and variance hasn't had time to compound. The BVAR's shrinkage introduces a small bias penalty.
- **Long-Term ($h = 12$):** BVAR dominates. Iterating a 111-parameter OLS matrix forward 12 times causes variance to explode. The Minnesota prior anchors the system, proving shrinkage is necessary for high-dimensional stability.
- **The Lasso NAs:** Why did BigVAR fail at $h = 6$? If L_1 sparsity deletes the wrong coefficients, the VAR characteristic roots fall inside the unit circle. The iterated forecast becomes unstable and collapses! *Diagnostic check: verify companion-matrix eigenvalues; compare iterated vs direct multi-step forecasts; test different penalties (group lasso) and stability-enforced routines.*

Lab Results: Error Distribution Analysis



Lab Results: Error Distribution Analysis

Takeaways: The Shape of Uncertainty

- **BVAR (Tight Density):** The Minnesota prior successfully anchors the 111-parameter matrix. Errors are highly concentrated near zero, proving shrinkage controls forecast variance.
- **OLS VAR (The Fat Tail):** Without regularization, iterating the matrix forward causes errors to compound exponentially. The long right tail visually represents the "Curse of Dimensionality" destroying the long-term forecast.
- **Lasso VAR (Truncation):** Performs decently, but structural instability (NAs) prevents it from maintaining a complete 24-month trajectory, highlighting the danger of algorithmic variable selection in time series.

Concluding Remarks

Project: 10-Minute Check-In

Our Standard Closing Questions

1. **What changed in OOS RMSE by horizon?**
2. **Which failure mode did we fix (or create)?**

Today's Expected Answers:

- **RMSE Shift:** OLS VAR(12) will likely perform catastrophically bad at $h = 12$ compared to the AR(12) from Lecture 1. BVAR and Sparse VAR will restore the RMSE to competitive levels.
- **Failure Mode Fixed:** We allowed variables to interact (system dynamics) without suffering from the massive parameter explosion inherent to VARs.
- **Failure Mode Created:** Shrinkage improves prediction but changes the implied dynamics; structural IRFs require explicit identification and sensitivity checks (and for sparse VARs, post-selection issues complicate inference).

Readings Discussion

Reading Discussion: Giannone, Lenza, & Primiceri (2015)

The Historical Critique of BVARs

For decades, frequentist econometricians criticized Bayesian VARs because setting the Minnesota Prior's "tightness" parameter (λ) felt dangerously subjective. How much should we shrink? Who decides?

The Bridge to Machine Learning:

- **Data-Driven Priors:** Giannone et al. prove that the shrinkage parameter λ does not have to be guessed. It can be rigorously estimated from the data by maximizing the *Marginal Likelihood*.
- **The ML Equivalence:** Maximizing the Bayesian Marginal Likelihood is asymptotically equivalent to minimizing **Leave-One-Out Cross-Validation (LOOCV)** error.
- **The Takeaway:** Bayesian prior selection and ML hyperparameter tuning are essentially the exact same mathematical operation! We are actively optimizing the Bias-Variance tradeoff.

Reading Discussion: Nicholson et al. (2020)

The Problem with "Blind" Lasso in Time Series

Standard Lasso (L_1 penalty) treats all features equally. If you feed it a VAR(12), it might keep lag 1 and lag 8 of Unemployment, but drop lags 2 through 7. This creates an erratic, economically nonsensical dynamic response.

Imposing Econometric Structure on ML:

- **Structured Penalties:** Nicholson et al. introduce the `BigVAR` framework, showing that ML models perform better when we force them to respect the architecture of time.
- **Group Lasso:** By grouping variables (e.g., "all lags of interest rates" or "all variables at lag 3"), the algorithm makes block decisions.
- **The Takeaway:** Pure algorithmic risk minimization often fails in macroeconomics if it ignores time-series logic. The best ML forecasts come from hybrid models that embed econometric structure into the penalty function.

Looking Ahead: Lecture 4

Next Session: Deep Learning for Sequences

So far, our models have been strictly linear (VAR, Ridge, Lasso). In Lecture 4, we abandon the linear paradigm entirely and introduce Deep Sequence Models (RNNs and LSTMs).

Topics to prepare for:

- Neural networks as compositions of functions (the Chain Rule).
- How Recurrent Neural Networks (RNNs) manage sequential memory.
- The "Black Box" critique and where econometric inference enters.

Reminder: Reading memos for Farrell et al. (2021) and Lim et al. (2021) are due before the next Lecture!